PT.SUMITOMO ELECTRIC WINTEC INDONESIA

$\bar{X}$ - R CONTROL CHART

No. Dokumen	4-QAS-PP07-001
Revisi	00
Tgl Berlaku	2-Dec-16

MACHINE	SECTION	OPERATION	DATE CONTROL LIMITS CALC.	SPECIFICATION	COURSE NO.	CONTROL ITEM (Δ) YES / NO
DF Tester (New)	QC	DF 250^0 C		min 12 max 40		O

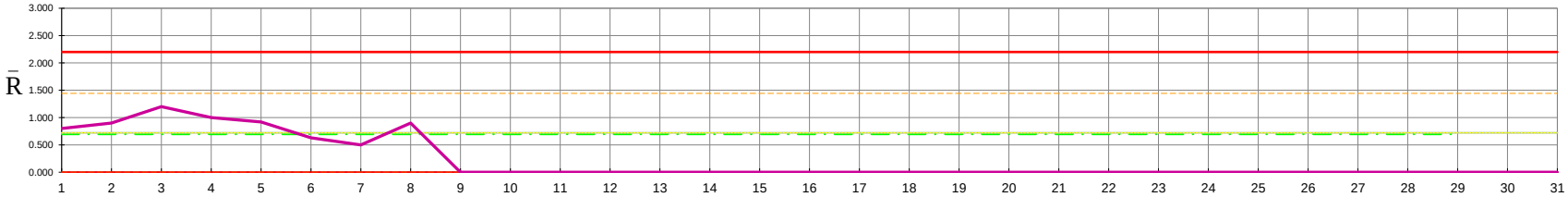
MONTH	CHARACTERISTIC	SAMPLE SIZE / FREQUENCY	TYPE / SIZE :	PIC:
May-25	Temperatur	n = 3	D2DSW2 / 0.50	DWI AS

X = AVERAGE $\bar{X}$ = 21.000 U.C.L. = $\bar{\bar{X}} + A_2\bar{R}$ = 24.0000 L.C.L. = $\bar{\bar{X}} - A_2\bar{R}$ = 18.0000

AVERAGE (X BAR CHART)

$\bar{R}$ = AVERAGE R = 0.700 U.C.L. = $D_4\bar{R}$ = 2.200 L.C.L. = $D_3\bar{R}$ = 0.00

RANGES (R - CHART)



Special Cause Marker Legend:
Test 1 ● Test 2 ◆ Test 3 ▲ Test 4 ■

SPECIAL CAUSES

- ☑ TEST 1: ANY POINT OUT OF CONTROL LIMIT
- ☑ TEST 2: 7 POINTS IN A ROW ON SAME SIZE OF CENTER LINE
- ☑ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
- ☑ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
- ☑ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
- ☐ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
- ☑ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
- ☑ TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

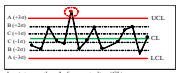
ACTION INSTRUCTION

1.
2.
3.
4.
5.

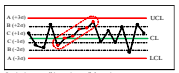
SUBGROUP SIZE	A_2	d_2	D_3	D_4
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

Special Cause

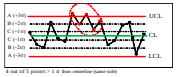
Titik berada di luar garis merah atau lebih dari 3 Ơ



6 titik berada dalam satu urutan secara terus menerus, baik posisi naik ataupun turun



4 atau 5 titik lebih dari 1Ơ dengan posisi yang sama dan titik keluar dari garis merah atau 3Ơ



Month																															
Date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
DATA 1	21.000	21.600	20.200	21.200	20.100	21.330	21.500	22.400																							
DATA 2	21.500	21.700	19.300	20.300	21.020	21.010	21.300	22.300																							
DATA 3	21.800	20.800	20.500	21.300	20.550	20.700	21.800	21.500																							
N	3	3	3	3	3	3	3	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AVERAGE, $\bar{X}$	21.433	21.367	20.000	20.933	20.557	21.013	21.533	22.067	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Range	0.800	0.900	1.200	1.000	0.920	0.630	0.500	0.900																							
Moving Range		0.002	0.768	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
SHIFT 1		DWI AS	DWI AS	DWI AS			DWI AS	DWI AS																							
SHIFT 2																															

CALCULATION

$\bar{\bar{X}}$ = 21.1129

$\bar{R}$ = 0.8562

s_{ST} = 0.5058 s_{LT} = 0.721

THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED

C_{PK} = MIN $\left[\frac{\bar{X} - LSL}{3s_{ST}}, \frac{USL - \bar{X}}{3s_{ST}}\right]$

= MIN [2.05 1.90]

= 1.90

C_p = $\left[\frac{USL - LSL}{6s_{ST}}\right]$

= [1.98]

CHECKED BY	APPROVED BY